

BrightLearn Business Analytics - Research Assignment 2

The Language of Business Analytics

01. Foundational Concepts of Analytics

1. Business Analytics

Definition:

Business Analytics is the use of data, statistical techniques and analytical methods to understand business performance and support better decisions. It connects data analysis to specific business problems such as improving profitability, managing risk or understanding customers. IBM describes business analytics as using statistical methods and computing technologies to uncover insights that support business decision-making.

Real-world example:

A bank analyses customer transaction data to identify customers who may be suitable for additional products such as credit cards or investment accounts.

Source: IBM - Business Analytics.

2. Data Analytics

Definition:

Data Analytics is the process of examining, cleaning, transforming and interpreting data to identify useful patterns and information. It is broader than simply producing reports because it seeks to turn

raw data into insights that can support decisions.

Real-world example:

A retailer analyses five years of sales data to identify which products sell most during different seasons.

Source: IBM - Business Analytics and Big Data Analytics.

3. Business Intelligence (BI)

Definition:

Business Intelligence refers to the technologies, systems and processes used to collect, organise, analyse and present business information. BI commonly focuses on reporting and monitoring current or historical performance, while more advanced analytics can be used to predict or recommend future actions.

Real-world example:

A bank uses a BI dashboard showing monthly revenue, expenses, customer numbers and sales performance by region.

Source: IBM - Business Intelligence.

4. Data-Driven Decision Making

Definition:

Data-driven decision making means using reliable evidence and analysis as an important basis for business decisions rather than relying only on intuition or assumptions. Good data-driven decisions also require managers to understand where the data came from and whether it is appropriate for the question being asked.

Real-world example:

A company uses customer complaints, sales data and customer surveys to decide whether it should change its product design.

Source: IBM; Harvard Business Review.

5. Big Data

Definition:

Big Data refers to extremely large, complex and rapidly generated datasets that can be difficult for traditional systems to store and analyse. It can involve different forms of data, including structured, semi-structured and unstructured information, and is commonly associated with characteristics such as volume, velocity and variety.

Real-world example:

A social-media company analyses millions of posts, images, videos and user interactions to understand customer behaviour.

Source: IBM - Big Data.

6. Descriptive Analytics

Definition:

Descriptive Analytics examines historical data to explain what happened. It summarises information using reports, tables, charts and dashboards so that an organisation can understand previous performance.

Real-world example:

A retailer produces a monthly report showing total sales, number of transactions and sales by product category.

Source: IBM.

7. Diagnostic Analytics

Definition:

Diagnostic Analytics investigates data to determine why something happened. It goes beyond reporting an outcome by looking for relationships, patterns and possible root causes.

Real-world example:

A company notices that sales fell by 15% and analyses customer data to determine whether the decline was caused by pricing, stock shortages or reduced customer traffic.

Source: IBM - Diagnostic Analytics.

8. Predictive Analytics

Definition:

Predictive Analytics uses historical and current data, statistical techniques and sometimes machine-learning models to estimate what is likely to happen in the future. It does not guarantee an outcome; instead, it provides an evidence-based forecast.

Real-world example:

An insurance company uses previous claims and customer information to estimate the probability that a policyholder will submit a claim.

Source: IBM - Predictive Analytics.

9. Prescriptive Analytics

Definition:

Prescriptive Analytics goes beyond predicting what may happen by suggesting what an organisation should do in response. It can use predictive models, optimisation techniques, constraints and business objectives to identify possible actions.

Real-world example:

A retailer's analytics system predicts increased demand for a product and recommends how much stock should be ordered to reduce the likelihood of shortages.

Source: IBM - Prescriptive Analytics.

10. Data Literacy

Definition:

Data Literacy is the ability to understand, interpret, question and communicate information presented in data. A data-literate employee should be able to understand basic measures, recognise misleading interpretations and use data appropriately when making decisions.

Real-world example:

A manager reviews a performance dashboard and is able to identify whether a percentage increase represents genuine growth or is simply caused by a change in the size of the customer base.

Source: BrightLearn assignment framework; IBM business analytics principles.

02. Data, Measurement & Reporting

11. Structured Data

Definition:

Structured Data is information organised according to a predefined format or schema, usually in rows and columns. Because its structure is consistent, it is relatively easy to store, search and analyse using databases and tools such as SQL.

Real-world example:

A bank's customer database containing customer numbers, account balances, transaction dates and transaction amounts is structured data.

Source: IBM - Structured vs Unstructured Data.

12. Unstructured Data

Definition:

Unstructured Data is information that does not follow a fixed database structure or predefined schema. Examples include documents, emails, photographs, videos, social-media posts and audio recordings.

Real-world example:

A bank analyses the text contained in customer complaints and call-centre recordings to identify common customer concerns.

Source: IBM - Structured vs Unstructured Data.

13. Data Warehouse

Definition:

A Data Warehouse is a centralised data environment designed primarily for reporting, analysis and business intelligence rather than day-to-day transaction processing. It normally brings information from multiple sources together and can retain historical data for analysis.

Real-world example:

A bank combines customer, loan, transaction and financial information in a data warehouse so analysts can produce management reports.

Source: Oracle - Data Warehousing Concepts.

14. Data Mining

Definition:

Data Mining involves examining large datasets to discover patterns, relationships, trends or other useful information that may not be immediately obvious. It can support activities such as customer analysis, fraud detection and prediction.

Real-world example:

A supermarket analyses millions of transactions and discovers that customers who buy baby products frequently purchase certain household products as well.

Source: IBM - Business Analytics and Big Data Analytics.

15. Data Visualization

Definition:

Data Visualization is the presentation of data through visual formats such as charts, graphs, maps and diagrams. Effective visualisation helps users identify trends, differences, relationships and unusual results more quickly than reviewing raw numbers alone.

Real-world example:

A sales department uses a line graph to show monthly revenue trends and a bar chart to compare sales between regions.

Source: IBM - Business Analytics.

16. Dashboard

Definition:

A Dashboard is a visual interface that brings important measures and performance information together in one place. Dashboards can provide regularly updated information so managers can monitor performance and identify issues requiring attention.

Real-world example:

A bank's executive dashboard displays revenue, customer growth, operational incidents, complaints and risk indicators.

Source: IBM - Business Intelligence and Business Analytics.

17. Metric

Definition:

A Metric is a measurable value used to describe or monitor an aspect of business performance. A metric does not necessarily represent a strategic priority; it becomes a KPI when it is specifically linked to an important business objective or critical success factor.

Real-world example:

A company's website page views may be a metric used to monitor online activity.

Source: IBM - Metrics and KPIs.

18. Key Performance Indicator (KPI)

Definition:

A KPI is a measurable indicator that tracks performance against an important business objective or critical success factor. Unlike an ordinary metric, a KPI is selected because it is particularly important to the success of the organisation and normally has a target or performance threshold.

Real-world example:

For a call centre, the percentage of calls answered within 60 seconds could be a KPI because customer service is a critical business objective.

Metric vs KPI: Every KPI is a metric, but not every metric is

a KPI. For example, website page views may be an ordinary metric, while conversion rate may be a KPI for an online retailer.

Source: IBM - Key Performance Indicators.

19. Benchmark

Definition:

A Benchmark is a reference point used to compare performance. A business may compare its results with previous performance, competitors, industry standards or an internally established target.

Real-world example:

A bank compares its customer complaint resolution time with the industry benchmark to determine whether its performance is in line with other organisations.

Source: BrightLeann assignment framework; IBM business performance principles.

20. Segmentation

Definition:

Segmentation means dividing a broad customer or business population into smaller groups based on shared characteristics. These characteristics may include demographics, behaviour, location, purchasing patterns or customer value.

Real-world example:

An online retailer divides customers into frequent buyers, occasional buyers and inactive customers and sends each group different marketing messages.

Source: IBM - Business Analytics.

03. Customer & Sales KPIs

21. Conversion Rate

Definition:

Conversion Rate measures the percentage of people who complete a desired action out of the relevant population. The action could be purchasing a product, completing an application, registering for an account or submitting a lead form.

Formula:

$$\text{Conversion Rate} = (\text{Number of conversions} \div \text{Number of relevant visitors/leads}) \times 100$$

Real-world example:

If 1,000 people visit an online store and 50 purchase something, the conversion rate is 5%.

Source: Google Analytics.

22. Customer Acquisition Cost (CAC)

Definition:

Customer Acquisition Cost measures the average amount a business spends to obtain a new customer. It normally considers relevant marketing and sales costs and divides them by the number of new customers acquired during the period.

Formula:

$$\text{CAC} = \text{Total acquisition costs} \div \text{Number of new customers acquired}$$

Real-world example:

If an online retailer spends R100,000 on sales and marketing and gains 500 new customers, its CAC is R200 per customer.

Source: HubSpot sales metrics.

23. Customer Lifetime Value (CLV/LTV)

Definition:

Customer Lifetime Value estimates the economic value a business expects to obtain from a customer over the duration of the relationship. It helps a business understand how much value customers generate relative to the cost of acquiring them.

Real-world example:

A subscription company estimates that an average customer generates R6,000 in contribution over their relationship with the company.

Source: HubSpot sales metrics.

24. Churn Rate

Definition:

Churn Rate measures the percentage of customers who stop using a company's product or service during a specified period. A high churn rate can indicate difficulties with customer satisfaction, pricing, service or competitive pressure.

Formula:

$$\text{Churn Rate} = \frac{\text{Customers lost during period}}{\text{Customers at beginning of period}} \times 100$$

Real-world example:

A subscription business starts the month with 10,000 customers and loses 300 during the month, resulting in a 3% monthly churn

rate.

Source: HubSpot sales metrics.

25. Customer Retention Rate

Definition:

Customer Retention Rate measures the percentage of existing customers that a business keeps over a specified period. It is closely related to churn because a higher retention rate generally corresponds to a lower customer loss rate.

Formula:

$$\text{Retention Rate} = \left[\frac{\text{Customers at end} - \text{New customers acquired}}{\text{Customers at beginning}} \right] \times 100$$

Real-world example:

A subscription company tracks how many customers remain subscribed after one year to assess the effectiveness of its retention initiatives.

Source: HubSpot sales metrics.

26. Net Promoter Score (NPS)

Definition:

Net Promoter Score measures customers' willingness to recommend a company, product or service to others. Customers normally provide a score from 0 to 10, with scores of 9-10 classified as Promoters, 7-8 as Passives and 0-6 as Detractors.

Formula:

$$\text{NPS} = \% \text{ Promoters} - \% \text{ Detractors}$$

Real-world example:

A bank asks customers how likely they are to recommend the bank and uses the resulting NPS to monitor customer experience.

Source: Qualtrics.

27. Average Order Value (AOV)

Definition:

Average Order Value measures the average amount of revenue generated each time a customer places an order. It helps businesses understand purchasing behaviour and evaluate strategies designed to increase the amount customers spend.

Formula:

$$\text{AOV} = \text{Total revenue} \div \text{Number of orders}$$

Real-world example:

An online retailer generates R500,000 from 2,000 orders, giving an AOV of R250.

Source: IBM KPI guidance; HubSpot sales metrics.

28. Sales Funnel

Definition:

A Sales Funnel describes the stages a potential customer moves through from becoming aware of a product or company to eventually making a purchase. Typical stages include awareness, interest, consideration, conversion and, in some models, retention or advocacy.

Real-world example:

An insurance company may track how many people see an advertisement, request a quote, submit an application and finally purchase a policy.

Source: BrightLearn assignment brief; business analytics and marketing measurement principles.

Important relationship: CLV and CAC

A business should consider CLV and CAC together rather than looking at customer acquisition cost in isolation. If acquiring customers is expensive relative to the value those customers generate, growth may not be financially sustainable.

For example, if a customer generates R6,000 of estimated lifetime value and costs R1,500 to acquire, the CLV:CAC ratio is 4:1.

04. Marketing & Campaign Analytics

29. Marketing Campaign

Definition:

A Marketing Campaign is a coordinated set of marketing activities designed to achieve a particular objective, such as increasing awareness, generating leads or increasing sales. A campaign normally has a defined target audience, message, channels, budget and measurement criteria.

Real-world example:

A retailer launches a four-week social-media and email campaign promoting its Black Friday products.

Source: BrightLearn assignment brief.

30. Post-Campaign Analysis

Definition:

Post-Campaign Analysis is the process of reviewing campaign results

after the campaign has run to determine what happened and what can be learned. It normally compares actual performance with campaign objectives and examines metrics such as conversions, revenue, cost and return.

Real-world example:

After an online advertising campaign, a company compares the number of leads generated, advertising costs and revenue generated with the original targets.

Source: BrightLearn assignment framework; Google Analytics campaign measurement.

31. Incremental Revenue

Definition:

Incremental Revenue is the additional revenue that can reasonably be attributed to an activity or campaign beyond what would have occurred without that activity. The key idea is to identify the extra revenue generated rather than simply counting all revenue occurring during the campaign.

Real-world example:

A retailer normally generates R1 million during a month but generates R1.2 million during a promotion; if analysis shows the promotion caused the additional R200,000, that amount represents incremental revenue.

Source: BrightLearn assignment framework; marketing attribution

principles.

32. Return on Investment (ROI)

Definition:

Return on Investment measures the financial return generated relative to the investment made. It is useful for determining whether an investment or activity generated sufficient financial benefit compared with its cost.

Formula:

$$ROI = (\text{Return} - \text{Investment Cost}) \div \text{Investment Cost} \times 100$$

Real-world example:

A company spends R100,000 on a campaign and generates R150,000 in profit attributable to the investment. The ROI calculation would compare the resulting return with the R100,000 investment.

Source: BrightLearn assignment brief; standard financial performance measurement.

33. Return on Ad Spend (ROAS)

Definition:

Return on Ad Spend measures the revenue generated for each unit of money spent specifically on advertising. Unlike broader ROI, ROAS focuses specifically on advertising revenue relative to advertising expenditure.

Formula:

$ROAS = \text{Revenue attributable to advertising} \div \text{Advertising spend}$

Real-world example:

If a company spends R50,000 on Google advertising and generates R200,000 in attributable revenue, its ROAS is 4:1.

Source: BrightLearn assignment brief; Google Analytics advertising measurement.

34. Click-Through Rate (CTR)

Definition:

Click-Through Rate measures the percentage of people who click on a digital advertisement, email link or other clickable marketing element after being exposed to it. It is commonly used to assess how effectively content encourages people to take the next action.

Formula:

$CTR = \text{Clicks} \div \text{Impressions} \times 100$

Real-world example:

An online advertisement receives 20,000 impressions and 400 clicks, giving a CTR of 2%.

Source: Google Analytics advertising and campaign measurement.

35. Bounce Rate

Definition:

Bounce Rate measures the percentage of website or app sessions that are classified as non-engaged sessions. In current Google Analytics terminology, it is the inverse of engagement rate, so the precise calculation depends on how an engaged session is defined.

Real-world example:

A visitor arrives on a webpage, does not engage with the site and leaves without meeting Google's criteria for an engaged session; that session contributes to bounce rate.

Source: Google Analytics Help.

36. A/B Testing

Definition:

A/B Testing is an experiment in which two versions of something, such as an advertisement, webpage or email, are compared to determine which version produces better results. Ideally, users are divided into appropriate groups so that differences in performance can be measured objectively.

Real-world example:

An online retailer sends two versions of an email, one with a "Buy Now" button and another with a "Shop Now" button, and compares their conversion rates.

Source: BrightLearn assignment framework; Google Analytics experimentation and measurement principles.

37. Lead Generation

Definition:

Lead Generation is the process of identifying and capturing people or organisations that may have an interest in a company's products or services. Leads can be captured through activities such as online forms, advertising, events, content downloads or enquiries.

Real-world example:

A financial services company runs an online advertisement offering a free financial assessment and collects contact details from people who complete the form.

Source: BrightLearn assignment framework; Google Analytics conversion measurement.

38. Marketing Attribution

Definition:

Marketing Attribution is the process of determining how credit for an important customer action should be allocated across the marketing touchpoints that influenced the customer's journey.

Attribution is important because customers may interact with several channels before completing a purchase or other key action.

Real-world example:

A customer first sees a company's social-media advertisement, later searches for the company on Google, receives an email and eventually makes a purchase. Attribution analysis helps the company understand the contribution of those different touchpoints.

Source: Google Analytics Help.

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